The New Kind of Convolution and Correlation Theorems Associated with the Linear Canonical Wavelet Transform

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***Abstract*—Linear canonical transform (LCT), distinguished by its three free parameters, provides remarkable flexibility, establishing itself as a fundamental tool for time-frequency analysis and the examination of non-stationary signals. In recent years, wavelet transform (WT) has gained substantial attention as a potent signal analysis technique. Nevertheless, researchers in the domains of signal processing and image processing are continuously striving to develop innovative techniques for a better understanding and analysis of diverse signals. This thesis focuses on the exploration of a novel transformation, namely linear canonical wavelet transform (LCWT), which seamlessly integrates the strengths of both LCT and WT while addressing their inherent limitations. LCWT has emerged as a robust tool for signal processing. However, the theoretical framework for certain aspects of this transformation, such as convolution and its correlation theorems, remains imperfect. In response, we propose a novel convolution method to enhance the understanding of LCWT. This paper begins with a concise introduction to the fundamental theory of LCWT. Subsequently, we introduce a pioneering convolution and correlation operator and derive the convolution and correlation theorem by amalgamating LCWT. Finally, leveraging the derived theorem, we have proposed the theory for a novel filtering design approach within the domain of LCWT.**

Keywords-Linear canonical wavelet transform; convolution theorem; correlation theorem

I. INTRODUCTION

Linear canonical transformation (LCT) represents a more generalized integral transformation. Both Fourier transform (FT) and fractional Fourier transform (FRFT)[1], which are familiar to us, can be viewed as special cases of LCT. With three free parameters, LCT generally offers higher flexibility compared to other transformations. However, due to its unsuitability for nonlinear relationships, it may lack the flexibility needed to effectively handle local features in the processing of certain signals or images.

Wavelet transform (WT) is a type of transformation that decomposes signals or images into frequency components of varying scales through wavelet decomposition. Building upon short-time Fourier transform (STFT)[2], WT introduces the concept of real-time variation in the window function, replacing the trigonometric function basis of traditional transforms with a decaying wavelet basis. This transformation performs exceptionally well in the analysis of non-stationary signals[3][9]. However, despite its advantages, WT has some limitations, including its inability to adequately consider both local and global characteristics and its limited ability to handle nonlinear signals.

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Therefore, researchers in related fields are dedicated to developing new techniques and methods, leading to the exploration of various transformation forms. Examples include fractional wavelet transform (FRWT)[4], linear canonical wavelet transform (LCWT)[5], among others. These transformations prove more effective in handling non-stationary signals, providing a more accurate reflection of their information and value[13]. By leveraging the advantages of both WT and LCT, LCWT demonstrates improved adaptability to different types of data. Simultaneously, it preserves the local characteristics of the signal while reducing correlation.

Convolution is a common mathematical operator and has been well applied and achieved good results in dealing with many mathematical problems[6][11]. Therefore, researchers continue to study the convolution correlation theorem and have obtained many theoretical results. After understanding the present situation, a new convolution operator for LCWT correlation theorem is proposed, which provides a new scheme for filter design in LCWT domain.

The article is structured as follows: Initially, we provide a concise overview of the background knowledge pertaining to LCWT and convolution, accompanied by the introduction of relevant lemmas. Subsequently, we derive a novel convolution and correlation theorem specific to LCWT. Finally, we put forth the theoretical method of an innovative filtering design approach within the domain of LCWT.

II. PRELIMINARIES

In this section, we will briefly introduce the necessary basic knowledge about LCWT and some convolution theorems to facilitate subsequent use.

Definition 2.1. Let $f \in L^2(R)$, wavelet transform of the function f is defined as[3]

$$W_f^{\psi}(a,b) = \mathcal{W}[f(t)](a,b) = \int_R f(t)\psi_{a,b}^*(t)\, dt \qquad (1)$$

where the function $\psi \in L^2(R)$ is a basic wavelet (wavelet base) or mother wavelet and satisfy the admissibility condition[9]

$$C_{\psi}=\int_{R}\frac{|\psi(\omega)|^{2}}{|\omega|}d\omega<\infty \quad (2)$$

and the basis function of the wavelet is defined as[11]

$$\psi_{a,b}(t)=\frac{1}{\sqrt{|a|}}\psi\left(\frac{t-b}{a}\right) \quad (3)$$

where $a,b\in R$ and $a\neq 0$, they represent the scale parameter and the shift parameter respectively.

The inverse transformation of the wavelet transformation is expressed as[10]

$$f(x)=\frac{1}{C_{\psi}}\iint_{\mathbb{R}^{2}}\frac{1}{a^{2}}W_{f}^{\psi}(a,b)\psi_{a,b}(x)dadb \quad (4)$$

Linear canonical wavelet transform (LCWT), as a generalized wavelet transform in the linear canonical domain, not only breaks through the limitation of WT analysis in time-Fourier domain, but also overcomes the limitation that LCT cannot characterize the local features of the signal.

Definition 2.2. Let $f(t)\in L^{2}(R)$, linear canonical wavelet transform with the parameter matrix $M=(A,B,C,D)$ and mother wavelet $\psi\in L^{2}(R)$ of a signal $f(t)\in L^{2}(R)$ is defined as[5]

$$LW_{f}^{M}(a,b)=\mathcal{LW}^{M}[f(t)](a,b)=\int_{\mathbb{R}}f(t)\psi_{M,a,b}^{*}(t)dt \quad (5)$$

The kernel function $\psi_{M,a,b}$ is a continuous wavelet function and satisfies[5]

$$\begin{aligned}\Psi_{M,a,b}(t)&=\exp\left(-\frac{jA(t^{2}-b^{2})}{2B}\right)\Psi_{a,b}(t)\\&=\frac{1}{\sqrt{|a|}}exp\left(-\frac{jA(t^{2}-b^{2})}{2B}\right)\Psi\left(\frac{t-b}{a}\right)\end{aligned} \quad (6)$$

where $a,b\in R$ and $a\neq 0$. When the parameter matrix $M=(0,1,-1,0)$, LCWT reduced to WT. When the parameter matrix $M=(cos\alpha,sin\alpha,-sin\alpha,cos\alpha)$, LCWT reduced to FRWT[4]

$$LW_{f,\alpha}^{M}(a,b)=\int_{R}f(t)\exp\left(-\frac{j}{2}(t^{2}-b^{2})\cot\alpha\right)\Psi_{M,a,b}^{*}(t)dt \quad (7)$$

At the same time, linear canonical wavelet transform also can be defined as[12]

$$LW_{f}^{M}(a,b)=\sqrt{2\pi a}\int_{\mathbb{R}}L_{f}^{M}(u)F_{\psi}^{*}\left(\frac{au}{B}\right)K_{M^{-1}}(b,u)du \quad (8)$$

where L_{f}^{M} and F_{ψ} are LCT operator and Fourier operator, and $K_{M^{-1}}$ is the kernel function of the inverse LCT transformation.

The inverse transformation of LCWT is expressed as[12]

$$f(x)=\frac{1}{2\pi C_{\psi}}\iint_{\mathbb{R}^{2}}\frac{1}{a^{2}}LW_{f}^{M}(a,b)\psi_{M,a,b}(x)dadb \quad (9)$$

At different scales, LCWT in the frequency domain can be conceived as a set of bandpass filters through which a signal passes in a series of linear regular domains. This underscores LCWT's advancement beyond the constraints of traditional WT in time-Fourier domain analysis, while also addressing the limitation of LCT in characterizing local signal features.

Definition 2.3 Let $f,h\in L^{2}(R)$ is the complex valued function, then convolution operations with only one variable are defined as[7]

$$g(\alpha,\beta)=(f\,\Theta_{1}\,h)(\alpha,\beta)=\int_{\mathbb{R}}f(\alpha,\tau)h(\alpha,\beta-\tau)d\tau \quad (10)$$

$$g(\alpha,\beta)=(f\otimes_{1}h)(\alpha,\beta)=\int_{\mathbb{R}}f(\alpha,\tau)h(\alpha,\beta+\tau)d\tau \quad (11)$$

III. New Kind Of Generalized Convolution Theorem And Correlation Theorem

Definition 3.1. For any two functions $f(t),g(t)\in L^{2}(R)$, we introduce the new generalized convolution operation $\oplus_{A_1,A_2,A_3}$ for LCT as follows:

$$\begin{aligned}\left(f\oplus_{A_1,A_2,A_3}h\right)(t)=\int_{\mathbb{R}}f(\tau)h(t-\tau)e^{-j\frac{a_1}{b_1}\tau\left(t+\frac{\tau}{2}\right)}\\ \phi_{a_1,a_2,a_3}(\mathrm{t},\tau)d\tau\end{aligned} \quad (12)$$

where the ϕ_{a_1,a_2,a_3} satisfy

$$\phi_{a_1,a_2,a_3}(\cdot,\diamond)=e^{j\frac{a_2}{2b_2}(\diamond)^2+j\left(\frac{a_1}{2b_1}-\frac{a_3}{2b_3}\right)(\cdot)^2} \quad (13)$$

is a domain-independent weighted function. And the A_1,A_2,A_3 are the LCT parameter matrix of $f(t),h(t),\left(f\oplus_{A_1,A_2,A_3}h\right)(t)$.

Theorem 3.2. (Convolution theorem) Let $\psi_f\in L^{2}(R)$ and $\psi_h\in L^{1}(R)\cap L^{2}(R)$ be two admissible wavelets, and let $LW_{f,A_3}^{\psi_f},LW_{h,A_3}^{\psi_h}$ indicate that the functions $f\in L^{2}(R)$ and $h\in L^{1}(R)\cap L^{2}(R)$ have linear canonical wavelet transform of the parent wavelets ψ_f and ψ_h, respectively, if g and ψ_g satisfy $g=\left[f\oplus_{A_1,A_2,A_3}h\right]$,$\psi_g=\left[\psi_f\oplus_{A_1,A_2,A_3}\psi_h\right]$,then

$$LW_{g,A_3}^{\psi_g}(a,b)=\frac{1}{\sqrt{a}}\left(LW_{f,A_3}^{\psi_f}\,\Theta_{1}LW_{h,A_3}^{\psi_h}\right)(a,b) \quad (14)$$

Proof. According to the Reference[8], if ψ_f and ψ_h are two admissible wavelets, then ψ_g is also an admissible wavelet. LCWT of the function $g(x)$ is given by (5), so we can get

$$LW_{g,A_3}^{\psi_g}(a,b)=\int_{R}g(x)\psi_{A_3,a,b}^{*}(x)\,dx \quad (15)$$

then, combining with the (3), (6) and (15), we obtain

$$\begin{aligned}LW_{g,A_3}^{\psi_g}(a,b)=\frac{1}{\sqrt{a}}\int_{\mathbb{R}}g(x)\psi_g^{*}\left(\frac{x-b}{a}\right)\\ \times exp\left(\frac{ja_3(x^2-b^2)}{2b_3}\right)dx\end{aligned} \quad (16)$$

Combining known and convolution operators (12), we get

$$g(x)=\left[f\oplus_{A_1,A_2,A_3}h\right](x)=\int_{\mathbb{R}}f(\tau)h(t-\tau)\times e^{-j\frac{a_1}{2b_1}[\tau(2x+\tau)-x^2]+j\frac{a_2}{2b_2}\tau^2-j\frac{a_3}{2b_3}x^2}d\tau \quad (17)$$

$$\Psi_g\left(\frac{x-b}{a}\right)=\left[\Psi_f\oplus_{A_1,A_2,A_3}\Psi_h\right]\left(\frac{x-b}{a}\right)=\int_{\mathbb{R}}\Psi_f(y)\Psi_h\left(\frac{x-b}{a}-y\right)\exp\{-j\frac{a_1}{2b_1}[y\left(\frac{2(x-b)}{a}+y\right)-\left(\frac{x-b}{a}\right)^2]+j\frac{a_2}{2b_2}y^2-j\frac{a_3}{2b_3}\left(\frac{x-b}{a}\right)^2\}dy \quad (18)$$

then, substituting formula (17), (18) into (16), we can obtain

$$LW_{g,A_3}^{\Psi_g}(a,b)=\frac{1}{\sqrt{a}}\iiint_{R^3}f(\tau)h(x-\tau)\Psi_f^*(y)\Psi_h^*(\frac{x-b}{a}-y)\exp\{-j\frac{a_1}{2b_1}\left[2\tau x+\tau^2-x^2-\frac{2y(x-b)}{a}-y^2+\left(\frac{x-b}{a}\right)^2\right]+j\frac{a_2}{2b_2}[\tau^2-y^2]+j\frac{a_3}{2b_3}\left[\left(\frac{x-b}{a}\right)^2-b^2\right]\}d\tau\,dy\,dx \quad (19)$$

Let $\lambda=x-\tau$ and $\mu=b+ay-\tau$, at the same time, satisfying $(\tau-b)\lambda=(1-a^2)\mu^2+(a^2-1)\mu b+\mu(\tau-\lambda)$, then (19) can be written as

$$\begin{aligned}LW_{g,A_3}^{\psi_g}(a,b)&=\frac{1}{\sqrt{a}}\iiint_{R^3}f(\tau)h(\lambda)\psi_f^*\left(\frac{\tau-(b-\mu)}{a}\right)\psi_h^*\left(\frac{\lambda-\mu}{a}\right)\times\exp\left(\frac{ja_3(\tau^2-(b-\mu)^2)+\lambda^2-\mu^2}{2b_3}\right)d\tau\frac{d\mu}{a}d\lambda\\&=\frac{1}{\sqrt{a}}\int_R\left[\frac{1}{\sqrt{a}}\int_R f(\tau)\psi_f^*\left(\frac{\tau-(b-\mu)}{a}\right)\times\exp\left(\frac{ja_3(\tau^2-(b-\mu)^2)}{2b_3}\right)d\tau\left[\frac{1}{\sqrt{a}}\int_R h(\lambda)\times\psi_h^*\left(\frac{\lambda-\mu}{a}\right)\exp\left(\frac{ja_3(\lambda^2-\mu^2)}{2b_3}\right)d\lambda d\mu\right.\right.\\&=\frac{1}{\sqrt{a}}\int_R LW_{f,A_3}^{\psi_f}(a,b-\mu)\times LW_{h,A_3}^{\psi_h}(a,\mu)d\mu\\&=\frac{1}{\sqrt{a}}\left(LW_{f,A_3}^{\psi_f}\Theta_1 LW_{h,A_3}^{\psi_h}\right)(a,b)\end{aligned} \quad (20)$$

The proof is completed.

Definition 3.3. Let the two functions $f(t),g(t)\in L^2(R)$, the new generalized convolution operation $\odot_{A_1,A_2,A_3}$ for LCT is defined as

$$\left(f\odot_{A_1,A_2,A_3}h\right)(t)=\int_{\mathbb{R}}f(\tau)h(t+\tau)e^{-j\frac{a_1}{b_1}\tau\left(t-\frac{\tau}{2}\right)}\phi_{a_1,a_2,a_3}(t,\tau)d\tau \quad (21)$$

where the ϕ_{a_1,a_2,a_3} satisfy

$$\phi_{a_1,a_2,a_3}(\cdot,\diamond)=e^{j\frac{a_2}{2b_2}(\diamond)^2+j(\frac{a_1}{2b_1}-\frac{a_3}{2b_3})(\cdot)^2} \quad (22)$$

is a domain-independent weighted function. The A_1,A_2,A_3 are the LCT parameter matrix of $f(t),h(t),\left(f\odot_{A_1,A_2,A_3}h\right)(t)$.

Theorem 3.4. (Correlation theorem) Let $\psi_f\in L^2(R)$ and $\psi_h\in L^1(R)\cap L^2(R)$ be two admissible wavelets, and let $LW_{f,A_3}^{\Psi_f},LW_{h,A_3}^{\Psi_h}$ indicate that the functions $f\in L^2(R)$ and $h\in L^1(R)\cap L^2(R)$ have linear canonical wavelet transform of the mother wavelets ψ_f and ψ_h, if g and ψ_g satisfy $g=\left[f\odot_{A_1,A_2,A_3}h\right]$,$\psi_g=\left[\psi_f\odot_{A_1,A_2,A_3}\psi_h\right]$,then

$$LW_{g,A_3}^{\psi_g}(a,b)=-\frac{1}{\sqrt{a}}\left(LW_{f,A_3}^{\psi_f}\otimes_1 LW_{h,A_3}^{\psi_h}\right)(a,-b) \quad (23)$$

Proof. If ψ_f and ψ_h are two admissible wavelets, then ψ_g is also an admissible wavelet[8]. LCWT of the function $g(x)$ is given by (5), so we can get

$$LW_{g,A_3}^{\psi_g}(a,b)=\int_{\mathbb{R}}g(x)\psi_{A_3,a,b}^*(x)dx \quad (24)$$

then, bring (6) into the (24), we can get

$$LW_{g,A_3}^{\psi_g}(a,b)=-\frac{1}{\sqrt{a}}\left(LW_{f,A_3}^{\psi_f}\odot_1 LW_{h,A_3}^{\psi_h}\right)(a,-b) \quad (25)$$

Combining known and convolution operators (21), we obtain

$$g(x)=\left[f\odot_{A_1,A_2,A_3}h\right](x)=\int_{\mathbb{R}}f(\tau)h(x+\tau)\times e^{-j\frac{a_1}{2b_1}[\tau(2x-\tau)-x^2]+j\frac{a_2}{2b_2}\tau^2-j\frac{a_3}{2b_3}x^2}d\tau \quad (26)$$

$$\Psi_g\left(\frac{x-b}{a}\right)=\left[\Psi_f\odot_{A_1,A_2,A_3}\Psi_h\right]\left(\frac{x-b}{a}\right)=\int_{\mathbb{R}}\Psi_f(y)\Psi_h\left(\frac{x-b}{a}+y\right)\exp\{j\frac{a_1}{2b_1}[y\left(y-\frac{2(x-b)}{a}\right)+\left(\frac{x-b}{a}\right)^2]+j\frac{a_2}{2b_2}y^2-j\frac{a_3}{2b_3}\left(\frac{x-b}{a}\right)^2\}dy \quad (27)$$

then, substituting (26), (27) into (25), we can get

$$LW_{g,A_3}^{\Psi_g}(a,b)=\frac{1}{\sqrt{a}}\iiint_{R^3}f(\tau)h(x+\tau)\Psi_f^*(y)\Psi_h^*(\frac{x-b}{a}+y)\exp\{j\frac{a_1}{2b_1}\left[\tau(\tau-2x)+x^2-y\left(y-\frac{2(x-b)}{a}\right)-\left(\frac{x-b}{a}\right)^2\right]+j\frac{a_2}{2b_2}[\tau^2-y^2]+j\frac{a_3}{2b_3}\left[\left(\frac{x-b}{a}\right)^2-b^2\right]\}d\tau\,dy\,dx \quad (28)$$

Let $\lambda=x+\tau$ and $\mu=\tau+b-ay$, at the same time, satisfying $(a^2+1)\mu^2+2a^2\tau^2=(a^2+1)\mu b+(2a^2-1)\lambda\tau+\lambda(\mu-b)$, then (28) can be written as

$$LW_{g,A_3}^{\Psi_g}(a,b) = \frac{1}{\sqrt{a}} \iiint_{R^3} f(\tau) h(\lambda) \Psi_f^*(\frac{\tau-(\mu-b)}{a}) \Psi_h^*(\frac{\lambda-\mu}{a})$$
$$\exp\left(\frac{ja_3(\tau^2-(\mu-b)^2+\lambda^2-\mu^2)}{2b_3}\right) d\tau \left(-\frac{d\mu}{a}\right) d\lambda$$
$$= -\frac{1}{\sqrt{a}} \int_{\mathbb{R}} [\frac{1}{\sqrt{a}} \int_{\mathbb{R}} f(\tau) \Psi_f^* \left(\frac{\tau-(\mu-b)}{a}\right)$$
$$\exp\left(\frac{ja_3(\tau^2-(\mu-b)^2)}{2b_3}\right) d\tau [\frac{1}{\sqrt{a}} \int_{\mathbb{R}} h(\lambda)$$
$$\Psi_h^* \left(\frac{\lambda-\mu}{a}\right) \exp\left(\frac{ja_3(\lambda^2-\mu^2)}{2b_3}\right) d\lambda d\mu$$
$$= -\frac{1}{\sqrt{a}} \int_{\mathbb{R}} LW_{f,A_3}^{\Psi_f}(a,\mu-b) LW_{h,A_3}^{\Psi_h}(a,\mu) d\mu$$
$$= -\frac{1}{\sqrt{a}} \left(LW_{f,A_3}^{\Psi_f} \otimes_1 LW_{h,A_3}^{\Psi_h}\right)(a,-b) \tag{29}$$

The proof is completed.

IV. Filter Design Idea For Lcwt With The Convolution

In signal processing, in order to collect relatively pure signals, it is necessary to de-noise by filtering. The operation of a filter is often described as a convolution operation, where the response of the filter can be viewed as the convolution of the input signal with the filter kernel. The theorem presented above demonstrates the independence of correlation operations in the linear canonical wavelet and convolution. Furthermore, the joint time-frequency representation resulting from the convolution of two signals can be expressed as the convolution of their joint time-frequency representations at a fixed frequency in time or space variables. This implies that filtering can be accomplished through the following methods, and the detailed process and steps are explained in Fig. 1 and Tab. 1, respectively.

Let $\psi_u, \psi_v \in L^1(R) \cap L^2(R)$ be two admissible wavelets, $N(a,b) \in L^2(R^2)$, and let $LW_{f,A_3}^{\Psi_u}$ indicate that the functions $f(x) \in L^2(R)$ have linear canonical wavelet transform of the mother wavelets ψ_u ,then

$$G(a,b) = LW_{f,A_3}^{\Psi_u}(a,b) N(a,b) \tag{30}$$

By inverting the linear canonical wavelet transform of (30), we can obtain

$$g(x) = \frac{1}{2\pi C_{\Psi_v}} \iint_{\mathbb{R}^2} \frac{1}{a^2} LW_{f,A_3}^{\Psi_u}(a,b) N(a,b) \Psi_v \left(\frac{x-b}{a}\right) da db$$
$$= \frac{1}{2\pi C_{\Psi_v}} \iint_{\mathbb{R}^2} \frac{1}{a^2} \frac{1}{\sqrt{a}} \int_{\mathbb{R}} f(\tau) \Psi_u^* \left(\frac{\tau-b}{a}\right) \exp\left(\frac{ja_3(\tau^2-b^2)}{2b_3}\right)$$
$$N(a,b) \Psi_v \left(\frac{x-b}{a}\right) da db \tag{31}$$

Next, we define the $h(x,\tau)$ as

$$h(x,\tau) = \frac{1}{2\pi C_{\Psi_v}} \iint_{\mathbb{R}^2} \frac{1}{a^{\frac{5}{2}}} \Psi_u^* \left(\frac{\tau-b}{a}\right) \exp\left(\frac{ja_3(\tau^2-b^2)}{2b_3}\right)$$
$$N(a,b) \Psi_v \left(\frac{x-b}{a}\right) da db \tag{32}$$

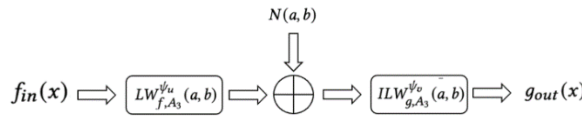

Figure 1. Flow chart for Spatial Variable Filtering in the LCWT Domain.

TABLE I. FILTER ASSUMPTION STEPS EXPLAINED.

Step number	**Operation and specific content explanation**
1	Input the original signal $f_{in}(x)$ into the filtering
2	$LW_{f,A_3}^{\Psi_u}$ is obtained by LCWT of the original signal.
3	Multiply $LW_{f,A_3}^{\Psi_u}$ by the transfer function $N(a,b)$[a].
4	Do inverse transformation of LCWT to $ILW_{f,A_3}^{\Psi_u}$.
5	Output filtered signal $g_{out}(x)$.

a. The transfer function $N(a,b)$ can be seen as a simplified expression of the convolution theorem.

From this, we obtain the space variable filter in LCWT domain

$$g(x) = \int_{\mathbb{R}} f(\tau) d\tau h(x,\tau) = (f \circledast h)(x) \tag{33}$$

Remark 1. $(f \circledast h)(x) = \int_R f(\tau) h(x,\tau) d\tau$ is the generalized convolution[7].

V. Conclusion

Expanding on LCWT, this paper delves into innovative correlation theorems involving convolution and combinations. It begins with a thorough review of background knowledge and correlation theorems linked to LCWT and WT. Subsequently, the paper introduces inventive convolution operators to derive new convolution theorems. Finally, leveraging the novel convolution theorem, the paper explores the theoretical design approach for spatial variable filtering methods within LCWT domain. In future theoretical research, we will further explore the knowledge associated with the new convolution-related theorems.

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